Notes on Fermi gas at finite $\tau \equiv k_B T$

Calculation of integrals with Fermi distribution function

Fermi-Dirac distribution $f(\varepsilon) = \dfrac{1}{e^{\frac{\varepsilon-\mu}{\tau}}+1}$

$$N = \underbrace{2}_{\text{spin degeneracy}} \int_0^\infty d\varepsilon \, \underbrace{D(\varepsilon)}_{\text{density of states}} f(\varepsilon)$$

Derive $D(\varepsilon)d\varepsilon$ from $D(k)\,dk = \dfrac{V}{2\pi^2} k^2 dk$:

$$\varepsilon = \frac{\hbar^2 k^2}{2m} \Rightarrow dk = \sqrt{\frac{2m}{\hbar^2}} \frac{d\varepsilon}{2\sqrt{\varepsilon}}$$

$$\Rightarrow \frac{V}{2\pi^2} \frac{2m\varepsilon}{\hbar^2} \sqrt{\frac{2m}{\hbar^2}} \frac{d\varepsilon}{2\sqrt{\varepsilon}} = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{\varepsilon}\, d\varepsilon = D(\varepsilon)d\varepsilon$$

in 3 dimensions.

Let us now represent N slightly differently:

$$N = \int_0^\infty d\varepsilon \left(\frac{d}{d\varepsilon} G(\varepsilon)\right) f(\varepsilon)$$

where $\dfrac{d}{d\varepsilon}G(\varepsilon) = 2D(\varepsilon) = \dfrac{V}{2\pi^2}\left(\dfrac{2m}{\hbar^2}\right)^{3/2}\sqrt{\varepsilon}$

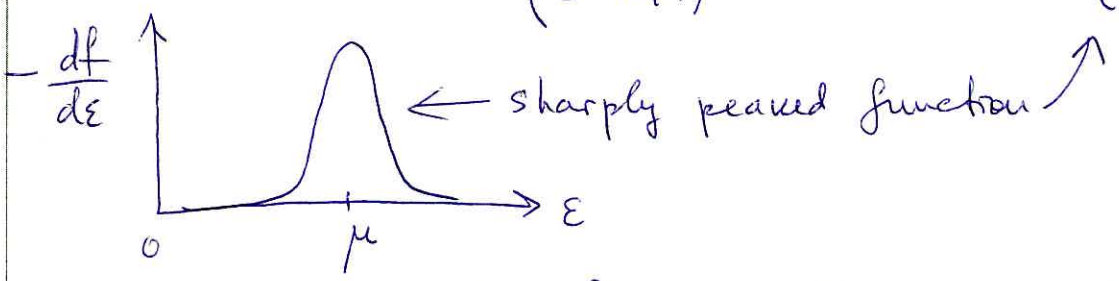
$$\Rightarrow G(\varepsilon) = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{3/2}$$

(meaning: number of states with energy less or equal to ε)

By integration by parts

$$N = \underbrace{G(\varepsilon) f(\varepsilon)\Big|_0^\infty}_{\left.\frac{\varepsilon^{3/2}}{e^{\frac{\varepsilon-\mu}{\tau}}+1}\right|_0^\infty = 0} - \int_0^\infty d\varepsilon \, G(\varepsilon) \frac{df}{d\varepsilon}$$

But $$-\frac{\partial f}{\partial \varepsilon} = \frac{1}{\tau}\,\frac{e^{\frac{\varepsilon-\mu}{\tau}}}{\left(e^{\frac{\varepsilon-\mu}{\tau}}+1\right)^2} = \frac{1}{\tau}\,\frac{1}{4\cosh^2\left(\frac{\varepsilon-\mu}{2\tau}\right)}$$



Clearly, $$\int_0^\infty d\varepsilon \left(-\frac{\partial f}{\partial \varepsilon}\right) = (-)\left[f(\infty) - f(0)\right] = \frac{1}{e^{-\frac{\mu}{\tau}}+1}$$

But $\frac{\mu}{\tau} \sim \frac{50{,}000K}{\text{few } 100K} \gg 1$ always! (for degenerate fermi gas)

$$\Rightarrow e^{-\frac{\mu}{\tau}} = 0.$$

Thus $-\frac{df}{d\varepsilon}$ is essentially delta-function of its argument: area under the curve is 1, and it is peaked at $\varepsilon=\mu$.

Then $$N = \int_0^\infty d\varepsilon\, G(\varepsilon)\left(-\frac{df}{d\varepsilon}\right) = \int d\varepsilon \left[G(\mu) + (\varepsilon-\mu)G'(\mu) + \frac{1}{2}(\varepsilon-\mu)^2 G''(\mu) + \dots\right]\left(-\frac{df}{d\varepsilon}\right)$$

because in comparison with very sharp $\left(-\frac{df}{d\varepsilon}\right)$, $G(\varepsilon)$ is very smooth function $\Rightarrow$ can be Taylor-expanded!

$$\Rightarrow N \cong G(\mu)\int_0^\infty d\varepsilon \left(-\frac{df}{d\varepsilon}\right) + G'(\mu)\int_0^\infty d\varepsilon\,(\varepsilon-\mu)\left(-\frac{df}{d\varepsilon}\right) + \frac{1}{2}G''(\mu)\int_0^\infty d\varepsilon\,(\varepsilon-\mu)^2\left(-\frac{df}{d\varepsilon}\right)$$

Consider $I_n = \int_0^\infty d\varepsilon\, (\varepsilon-\mu)^n \left(-\frac{df}{d\varepsilon}\right) = \left(x = \frac{\varepsilon-\mu}{\tau}\right) =$

$$= \int_{-\mu/\tau}^{\infty} dx \cdot \tau\, (\tau x)^n \frac{1}{\tau} \frac{1}{4\cosh^2\left(\frac{x}{2}\right)} = \left(-\frac{\mu}{\tau} \to -\infty\right) =$$

$$= \frac{\tau^n}{4} \int_{-\infty}^{\infty} dx \frac{x^n}{\cosh^2\left(\frac{x}{2}\right)}$$

Thus $I_n = 0$ for $n = 1, 3, 5 \ldots$ (odd).

Need to Find I_2 next $(n=2)$: (use Mathematica, for example)

$$I_2 = \frac{\tau^2}{4} \int_{-\infty}^{\infty} dx \frac{x^2}{\cosh^2\frac{x}{2}} = \tau^2 \int_{-\infty}^{\infty} dx \frac{x^2}{\left(e^{\frac{x}{2}} + e^{-\frac{x}{2}}\right)^2} =$$

$$= \tau^2 \int_{-\infty}^{\infty} dx\, x^2 \frac{e^x}{(e^x+1)^2} = \tau^2 \left(\frac{\pi^2}{3}\right)$$ [~~see eq. 33 in the next book, p.4~~]

Hence $N = G(\mu) + \frac{1}{2} G''(\mu)\, I_2 =$

$$= \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \mu^{3/2} + \frac{1}{2} \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \frac{1}{\mu^{1/2}} \frac{\tau^2\pi^2}{3}$$

Let first $\underline{\tau = 0}$: $N = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \mu(\tau=0)^{3/2} = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon_F^{3/2}$

$$\Rightarrow \mu(\tau=0) \equiv \varepsilon_F = \frac{\hbar^2}{2m}\left(3\pi^2 \frac{N}{V}\right)^{2/3}$$

(Fermi Energy)

Now finite $\underline{\tau \neq 0}$: $\mu(\tau)$ is found from

$$N = \frac{N}{\varepsilon_F^{3/2}} \mu^{3/2} + \frac{\pi^2\tau^2}{8} \frac{N}{\varepsilon_F^{3/2}\mu^{1/2}}$$

divide by N: $1 = \left(\frac{\mu}{\varepsilon_F}\right)^{3/2} + \frac{\pi^2\tau^2}{8\,\varepsilon_F^{3/2}\mu^{1/2}}$

$$\frac{\mu}{\varepsilon_F} = \left(1 - \frac{\pi^2 \tau^2}{8\,\varepsilon_F^{3/2}\,\mu^{1/2}}\right)^{2/3} = 1 - \frac{\pi^2 \tau^2}{12\,\varepsilon_F^2}$$

replace $\mu \to \varepsilon_F$ here because the difference contributes only in $O(\tau^4)$ terms which are too small to be considered here

remember that $\frac{\tau}{\varepsilon_F} \ll 1$, $\frac{\tau}{\mu} \ll 1$

$$\Rightarrow \mu(\tau) = \varepsilon_F \left(1 - \frac{\pi^2}{12}\left(\frac{\tau}{\varepsilon_F}\right)^2\right) \qquad \text{(Very weak } \tau\text{-dependence)}$$

<u>Internal energy</u> can be analyzed similarly:

$$U = \underset{\text{spin}}{(2)} \int_0^\infty d\varepsilon\, D(\varepsilon)\, f(\varepsilon)\, \varepsilon = \int_0^\infty d\varepsilon\, \frac{d\tilde{G}}{d\varepsilon}\, f(\varepsilon)$$

By definition $\frac{d}{d\varepsilon}\tilde{G}(\varepsilon) = 2\,D(\varepsilon)\,\varepsilon = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{3/2} =$

$$= \frac{3}{2}\underbrace{\frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}}_{\frac{N}{\varepsilon_F^{3/2}}} \varepsilon^{3/2} = \frac{3}{2} N \left(\frac{\varepsilon}{\varepsilon_F}\right)^{3/2}$$

$$\Rightarrow \tilde{G}(\varepsilon) = \frac{3}{5}\frac{N}{\varepsilon_F^{3/2}}\,\varepsilon^{5/2} \quad ; \quad \tilde{G}''(\varepsilon) = \frac{9}{4} N \frac{\varepsilon^{1/2}}{\varepsilon_F^{3/2}}$$

$$U(\tau) = \underbrace{\tilde{G}(\varepsilon) f(\varepsilon)\Big|_0^\infty}_{\left.\frac{\varepsilon^{5/2}}{e^{\frac{\varepsilon-\mu}{\tau}}+1}\right|_0^\infty = 0} + \int_0^\infty d\varepsilon\, \tilde{G}(\varepsilon)\left(-\frac{df}{d\varepsilon}\right) =$$

expand smooth function $\tilde{G}(\varepsilon)$ about $\varepsilon = \mu$:

$$= \int_0^\infty d\varepsilon \left[\tilde{G}(\mu) + (\varepsilon-\mu)\tilde{G}'(\mu) + \frac{1}{2}(\varepsilon-\mu)^2\,\tilde{G}''(\mu)\right]\left(-\frac{df}{d\varepsilon}\right)$$

$$U(\tau) = \tilde{G}(\mu) \underbrace{\int_0^\infty d\varepsilon \left(-\frac{df}{d\varepsilon}\right)}_{I_0 = 1} + \tilde{G}'(\mu) \underbrace{\int_0^\infty d\varepsilon (\varepsilon-\mu)\left(-\frac{df}{d\varepsilon}\right)}_{I_1 = 0} +$$

$$+ \frac{1}{2} \tilde{G}''(\mu) \underbrace{\int_0^\infty d\varepsilon \, (\varepsilon-\mu)^2 \left(-\frac{df}{d\varepsilon}\right)}_{I_2 = \frac{\pi^2\tau^2}{3}}$$

$$U(\tau) = \frac{3}{5} N \frac{\mu^{5/2}}{\varepsilon_F^{3/2}} + \frac{1}{2} \frac{\pi^2\tau^2}{3} \frac{9}{4} N \frac{\mu^{1/2}}{\varepsilon_F^{3/2}} =$$

$$\cong \frac{3}{5} N \frac{\left(\varepsilon_F \left(1 - \frac{\pi^2\tau^2}{12\varepsilon_F^2}\right)\right)^{5/2}}{\varepsilon_F^{3/2}} + \frac{3\pi^2\tau^2}{8} N \frac{\varepsilon_F^{1/2}}{\varepsilon_F^{3/2}} =$$

$$= \frac{3}{5} N \varepsilon_F \left(1 - \frac{5}{2} \frac{\pi^2\tau^2}{12\varepsilon_F^2}\right) + \frac{3}{8} N \frac{\pi^2\tau^2}{\varepsilon_F} =$$

$$= \frac{3}{5} N \varepsilon_F - \frac{1}{8} \frac{\pi^2\tau^2}{\varepsilon_F^2} N\varepsilon_F + \frac{3}{8} \frac{\pi^2\tau^2}{\varepsilon_F^2} N\varepsilon_F =$$

$$= \underbrace{\frac{3}{5} N\varepsilon_F}_{\text{zero temperature result}} + \underbrace{N\varepsilon_F \frac{\pi^2\tau^2}{4\varepsilon_F^2}}_{\text{finite } \tau \text{ correction}}$$

Thus $C_V(\tau) = \left(\frac{\partial U}{\partial \tau}\right)_V = N\varepsilon_F \frac{\pi^2\tau}{2\varepsilon_F^2} = \frac{\pi^2}{2} N \frac{\tau}{\varepsilon_F}$

linear in T specific heat!

<u>Entropy</u> $C_V = \tau \left(\frac{\partial \sigma}{\partial \tau}\right)_V \Rightarrow \left(\frac{\partial \sigma}{\partial \tau}\right)_V = \frac{\pi^2 N}{2\varepsilon_F}$

$$\Rightarrow \sigma(\tau) = \frac{\pi^2 N}{2} \frac{\tau}{\varepsilon_F}$$